

# Mostafa Ashraf Suliman

**Computer Science Student | Aspiring Machine Learning Engineer**

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<https://github.com/Mostafa-Ashraf-1> |

<https://mostafa-ashraf-1.github.io/Portfolio/>

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## Profile

Third-year **Computer Science student** with strong foundations in **machine learning, algorithms, and software systems**. Experienced in implementing ML models **from scratch** and with frameworks, with a focus on **model evaluation and real-world problem solving**. Seeking internship opportunities in **machine learning or software engineering** to apply theoretical knowledge in practical environments.

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## Technical Skills

- **Programming:** Python, Java, C, HTML, CSS, JavaScript
  - **Machine Learning:** Regression, Classification (Binary, Multiclass, Multilabel), Model Evaluation, Threshold Tuning
  - **Deep Learning:** Neural Networks (from scratch), Keras; Introductory exposure to CNNs, RNNs, Transformers
  - **CS Fundamentals:** Algorithms & Data Structures, Computer Architecture, Operating Systems, Databases
  - **Systems:** Real-Time Systems (ESP32), Basic Networking
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## Experience

### Machine Learning Projects

#### *Academic & Personal*

- Implemented **regression and classification models from scratch** in Python to understand optimization, loss functions, and decision boundaries, then reimplemented them using **Keras** for experimentation.

- Developed a **Stroke Prediction classification project**, handling preprocessing, class imbalance, recall-focused threshold tuning, and evaluation using confusion matrices and ROC–AUC.
- Built **neural networks from scratch** (forward and backpropagation), followed by framework-based implementations.

## Software & Systems Projects

### *Academic*

- Designed and implemented a **real-time car parking system using ESP32**, addressing sensor integration and timing constraints.
- Simulated a **pipelined processor** demonstrating instruction flow and **data hazard handling**.
- Implemented a **Tic-Tac-Toe AI** using **Alpha-Beta Pruning**.
- Developed a **hotel reservation system (Java)**, **vaccination center reservation app (Python)**, and a **clinic website (HTML/CSS/JS)**.

## IT Intern

### **SCAD**

- Supported internal systems and end users, gaining hands-on experience with operating systems, basic networking, and structured troubleshooting.
- Collaborated within a professional technical environment and documented solutions.

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## Education

### **Bachelor of Computer Science (In Progress)**

Helwan National University, Cairo

Expected Graduation: 2027

**GPA:** 3.3 / 4.0

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